

Lecture 18

CONSTRAINT

SATISFACTION PROBLEMS

Artificial Intelligence
COSC-3112

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Today's Agenda

- Variations on CSP Formalism
- Variable Types
- Constraint Types
- Convert n-ary constraint to binary
- Inference in CSP
- Local Consistency
 - Node Consistency
 - Arc Consistency
 - Path Consistency
 - K- Consistency

Variations on CSP Formalism

Variable Types

- Simplest CSPs define variables that have **discrete, finite domains**.
 - E.g. map coloring, 8-queens.
- Discrete domains can be **infinite**
 - e.g. set of integers or strings.
 - cannot enumerate all possible combinations of values
 - Thus need to use a constraint language.

Variations on CSP Formalism

Variable Types

- E.g., for a job scheduling problem,
 - might have $T_1 + d_1 \leq T_2$
 - where T_1 and T_2 are start times for two tasks and
 - d_1 is the duration of the first one
 - This describe that the first task must finish before the second one starts

Variations on CSP Formalism

Variable Types

- CSPs with **continuous domains** are common in the real world and are studied in operations research.
 - E.g., scheduling experiments on Hubble Telescope requires meeting astronomical, precedence, and power constraints.
- Best-known category of continuous-domain CSPs is **linear programming problems**
 - where all constraints are linear equalities or inequalities.

Variations on CSP Formalism

Constraint Types

- The simplest type is the **unary constraint**, which restricts the value of a single variable:
 - $\langle (SA), SA \neq \text{green} \rangle$
- A **binary constraint** relates two variables:
 - $\langle (SA, NSW), SA \neq NSW \rangle$
 - A binary CSP is one with only binary constraints; it can be represented as a constraint graph.
- **Higher-order** constraints are possible.
 - E.g. Between (X, Y, Z) that specifies that Y is between X and Z

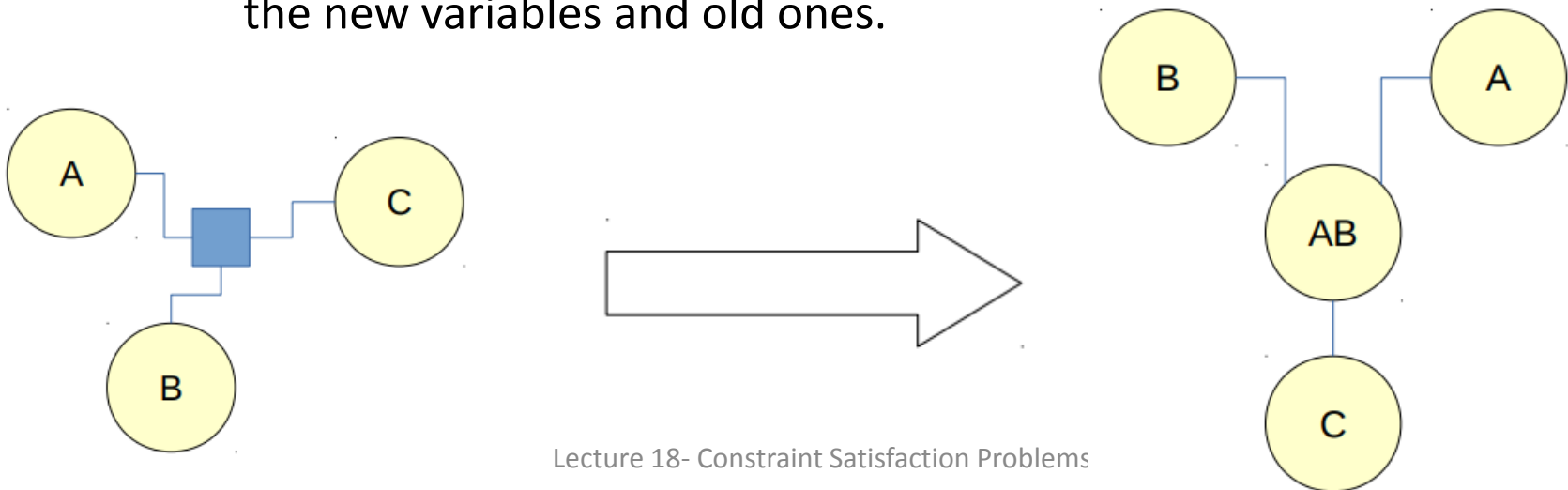
Variations on CSP Formalism

Constraint Types

- A constraint involving an arbitrary number of variables is called a **global constraint**.
 - One of the most common global constraints is *Alldiff*.
- Many real-world CSPs include **preference constraints** indicating which solutions are preferred:
 - constraint optimization problem (COP)

Convert n-ary constraint to binary one

- Example: How a single ternary constraint such as “ $A + B = C$ ” can be turned into three binary constraints?
 - Using an auxiliary variable:
 - First introduce a new variable AB and its domain.
 - Then create appropriate relations between the new variables and old ones.



Inference in CSP

- In regular state-space search, an algorithm can do only one thing: search.
- In CSPs there is a choice:
 - an algorithm can search (choose a new variable assignment from several possibilities) or
 - do a specific type of inference called **constraint propagation**
- Constraint Propagation
 - using the constraints to reduce the number of legal values for a variable,
 - which in turn can reduce the legal values for another variable, and so on (**Local Consistency**)

Local Consistency

- Node consistency
- Arc consistency
- Path consistency
- k-consistency

Local Consistency

Node Consistency

- A single variable is node-consistent if all the values in the variable's domain satisfy the variable's **unary constraints**.
- A network is node-consistent if **every** variable in the network is node-consistent.
- It can be done as a preprocessing step:
 - eliminate all inconsistent values from variables' domains.

Local Consistency

Arc Consistency

- A variable in a CSP is arc-consistent if **every value** in its domain satisfies the variable's binary constraints.
- More formally:
 - X_i is arc-consistent with respect to another variable X_j if
 - for every value in the current domain D_i there is some value in the domain D_j that satisfies the binary constraint on the arc (X_i, X_j) .
- A network is arc-consistent if every variable is arc consistent with every other variable.

Local Consistency

Path Consistency

- For many kind of problems, arc consistency fails to make enough inferences. Consider the map-coloring problem on Australia, but with only two colors allowed, red and blue...



Local Consistency

Path Consistency

- A two-variable set $\{X_i, X_j\}$ is path-consistent with respect to a third variable X_m if, for every assignment $\{X_i = a, X_j = b\}$ consistent with the constraints on $\{X_i, X_j\}$, there is an assignment to X_m that satisfies the constraints on $\{X_i, X_m\}$ and $\{X_m, X_j\}$
- This is called path consistency because one can think of it as looking at a path from X_i to X_j with X_m in the middle.

Local Consistency

Path Consistency (Example)

- We will make the set $\{WA, SA\}$ path consistent with respect to NT .
- We start by enumerating the consistent assignments to the set.
 - In this case, there are only two: $\{WA = \text{red} , SA = \text{blue}\}$ and $\{WA = \text{blue}, SA = \text{red}\}$.
- We can see that with both of these assignments NT can be neither red nor blue.
 - We eliminate both assignments, and we end up with no valid assignments for $\{WA, SA\}$.

Local Consistency

K Consistency

- A CSP is k -consistent if, for any set of $k - 1$ variables and for any consistent assignment to those variables
- A consistent value can always be assigned to any k th variable.
 - 1-consistency: Node consistency.
 - 2-consistency: Arc consistency.
 - 3-consistency: Path consistency.
- A CSP is strongly k -consistent if it is k -consistent and is also $(k-1)$ -consistent, $(k-2)$ -consistent, . . . all the way down to 1-consistent.

Local Consistency

Find a Solution

- Suppose there is a CSP with n nodes and strongly n -consistent. Now a solution can be found in this way:
 - Choose a consistent value for X_1 .
 - It is guaranteed to be able to choose a value for X_2 because the graph is 2-consistent, for X_3 because it is 3-consistent, and so on.
- It is guaranteed to find a solution in time $O(n^2d)$:
 - For each variable X_i , the algorithm only searches through the d values in the domain to find a value consistent with $X_1, \dots, X_{i-1}$.
- Any algorithm for establishing n -consistency must take time exponential in n in the worst case.